

JUSTIN KIM

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Education

University of Wisconsin–Madison | GPA: 3.76

Expected Graduation: Dec 2026

Bachelor of Science in Computer Science & Data Science

Madison, WI

- **Relevant Coursework:** Algorithms, Artificial Intelligence, Big Data Systems, Database Management Systems, Computer Networks, Machine Organization, Data Science Modeling I & II, Linear Algebra, Discrete Mathematics

Work Experience

Capital One

June 2026 – Aug 2026

Software Engineer Intern

Chicago, IL

- Built an AI-assisted authoring system that enables intent engineers to generate credit card fulfillment workflow configs from plain-English requirements, cutting a 2+ day manual process to under an hour.
- Designed a GraphRAG pipeline that grounded LLM output in approved workflow actions and validated each config against its schema, reaching 88% required-field completeness across 70 generated configs.
- Engineered the end-to-end React interface and AWS Lambda/DynamoDB backend for multi-turn workflow refinement, supporting iterative generation, validation, and revision of workflow configs.

MadLease

Jan 2024 – Present

Co-Founder & Founding Engineer

Madison, WI

- Founded and grew a UW–Madison sublease marketplace to 250+ registered accounts and 100+ listings.
- Architected the React/Spring Boot platform for authentication, messaging, listing management, and search, including a custom ranking system across rent, campus proximity, and 8 user-selectable filters.
- Launched a partner program for apartment managers with listing analytics and marketing visibility across 5+ properties.

Entegral

May 2025 – Aug 2025

Full Stack Software Engineer Intern

Madison, WI

- Developed a configurable crash-capture and insurance-intake platform in Angular and Spring Boot, allowing providers to define required documentation for each claim workflow.
- Created a reusable OCR module for VIN, license plate, and odometer extraction at 90%+ accuracy, now used company-wide as a standalone component.

Denari

Feb 2025 – May 2025

Software Engineer, Capstone Project

Madison, WI

- Built a citation-grounded RAG pipeline for tax document search, improving answer accuracy by 8% over a GPT-4o-mini baseline across 2,200+ CPA exam benchmark questions using OpenAI embeddings, TimescaleDB/pgvector, and AWS S3.

Astro AI Trading

Jan 2024 – Jan 2025

Machine Learning Engineer

Remote

- Led a 3-engineer team building stock forecasting models that combined market data with sentiment from 60+ daily news and social sources, reducing held-out RMSE by 32% over a price-only baseline.
- Clustered news sources by market sector using a BERT/PCA/KMeans pipeline, producing sector-specific sentiment features for downstream forecasting models.

Projects

QuizClash | *Devpost* | *Source Code*

NodeJS | **React** | **WebSocket** | **AWS**

- Created a real-time multiplayer quiz app converting PDFs and YouTube videos into Kahoot-style games with synchronized WebSocket gameplay and public/private rooms.
- Automated quiz creation with an ML pipeline for transcription, content extraction, question generation, and distractor selection using OCR, Whisper, T5, RoBERTa, and Sense2Vec, reaching 75% manually evaluated relevance.

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, Go, C, SQL, HTML/CSS

Libraries/Frameworks: React, Spring Boot, Node/Express, Flask, PyTorch, Angular, Scikit-Learn

Tools: Git, Docker, AWS (S3, EC2, Lambda, CloudFront, IAM, CDK), PostgreSQL, TimescaleDB/pgvector, Firebase